

# Jennifer Yang

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## EDUCATION

### Carnegie Mellon University

Master of Science in Robotics — GPA: 3.93/4.00

Pittsburgh, PA

Aug 2024

*Thesis Title:* DeltaWalker: A Soft, Linearly Actuated Delta Quadruped Robot

Bachelor of Science in Mechanical Engineering — GPA: 3.92/4.00

May 2023

## EXPERIENCE

### Research Assistant

May 2023 – Aug 2024

Carnegie Mellon University Zoom Lab

Pittsburgh, PA

- Built an affordable, easy-to-manufacture, omnidirectional quadruped robot with 3-DOF linear Delta actuators.
- Developed PyBullet simulations to test walking gaits, enabling rapid qualitative gait assessments.
- Applied nonlinear trajectory optimization to generate 4 different gaits with 4+ variations, and 1 rotation gait.
- Characterized robot performance with 50+ experiments, using computer vision to analyze position errors.
- Authored a comprehensive master's thesis and defended it in a formal presentation.

### Mechanical Engineer Intern

May 2022 – May 2023

Gecko Robotics

Pittsburgh, PA

- Designed, prototyped, and assembled an End-of-Line tester for the TOKA 4 Robot Center Body, streamlining build and verification processes by reducing bottlenecks and saving 200+ hours throughout the build cycle.
- Conducted thermal testing to assess system reliability and prevent overheating during extended operational cycles.
- Collaborated with end-users to refine testing unit design, incorporating feedback to optimize ergonomics and functionality through rapid prototyping techniques.

### Resident Assistant

Aug 2021 – May 2023

Carnegie Mellon University

Pittsburgh, PA

- Fostered community engagement among 40+ residents by organizing events, leading House Council initiatives, and mediating conflicts to maintain a positive living environment.

### Research Experiences for Undergraduates Student Intern

Jun 2021 – Aug 2021

Oregon State University Laboratory for Robotics and Applied Mechanics

Corvallis, OR

- Refined the sidewinding gait of a soft, pneumatic robot snake by assessing 2 control schemes and 8 flow rates.
- Enhanced understanding of its capabilities by evaluating performance across 3 terrains with 3 robot materials.

## LEADERSHIP

### Society of Women Engineers | Carnegie Mellon University

Aug 2019 – Sep 2022

- Coordinated the 2020 – 2022 STEM Career Fairs, managing catering for 200+ people, recruiting companies to attend, and partnering with CMU to promote the event.

## PROJECTS

### Haptic Mouse | Carnegie Mellon University

Mar 2024 – May 2024

- Worked with 4 peers to implement haptic feedback on a computer mouse using a rotary Delta actuator.
- Conducted user studies to refine haptic feedback for improving learning experiences for visually impaired users.

### PosChair: A Posture Enhancing Chair | Carnegie Mellon University

Aug 2022 – Dec 2022

- Worked with 4 peers to design and prototype a chair with an integrated sensor system for posture detection, haptic feedback for posture correction, and a GUI for visual feedback.

### Bellcranks and Dampers System Lead | Carnegie Mellon Racing

Aug 2021 – May 2022

- Designed and optimized the bellcranks and dampers system, ensuring it met specified design goals through kinematic simulations and stress analysis (FEA and hand calculations).
- CNC-machined components with tight tolerances and assembled the system onto the racecar.
- Documented design process and lessons learned in Confluence and the competition Design Binder.

## SKILLS

**Programming:** Python, MATLAB, C, C++, SQL, Julia, Arduino, PyBullet

**Software:** CAD (SolidWorks), CAM (HSMWorks, MasterCAM), FEA (SolidWorks FEA, ANSYS)

**Machines:** CNC Mill, Vertical Mill, Lathe, 3D Printer, Laser Cutter

**Languages:** Mandarin (Fluent speaker), French (Conversant)